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Thermodynamic assessment of the RE-B (RE=Ho, Er, Tm) binary systems

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ABSTRACT

To develop novel Nd-Fe-B-based permanent magnets with rare earth (RE) metals, phase equilibria and thermodynamic information of multi-component RE-Fe-B-based alloy systems is indispensable. In this work, thermodynamic assessment of the RE-B (RE=Ho, Er, Tm) binary systems as the important binary systems in the RE-Fe-B-based alloy systems were carried out using the CALPHAD (Calculation of Phase Diagram) method. The solution phases including liquid, hcp(α-Ho, α-Er and α-Tm) and rhombohedral(β-B) in the RE-B (RE=Ho, Er, Tm) binary systems are modeled by the substitutional solution model. The intermetallic compounds, HoB_2, HoB_4, HoB_6, HoB_{12}, HoB_{66}, ErB_2, ErB_4, ErB_{12}, ErB_{66}, TmB_2, TmB_4, TmB_{12} and $TmB_{66,}$ are treated as the stoichiometric compounds. Self-consistent thermodynamic parameters to describe Gibbs energies of phases in the RE-B (RE=Ho, Er, Tm) binary systems were obtained finally. The calculated results agree well with the reported data.

1. Introduction

Nd-Fe-B-based permanent magnets are widely applied to energy, automatic control, communications, medical and health, military and other industrial fields due to their excellent magnetic properties [1–5]. In order to further improve the magnetic performance of Nd-Fe-B-based magnets, high-performance Nd-Fe-B-based magnets currently need to be doped with an appropriate amount of heavy rare earth (RE) metals Dy or Tb [6–12]. Due to similar physical and chemical properties of rare earth metals, it is very worthwhile to explore outstanding magnetic properties of Nd-Fe-B-based magnets with other heavy rare earth metals (e.g. Ho, Er, Tm). The microstructure, magnetic properties and thermal stability of Nd-Fe-B-based magnets with the addition of Ho and Tm were investigated experimentally [13–18]. In particular, it was found that the addition of Ho in Nd-Fe-B-based magnets could promote the growth of $Nd_2Fe_{14}B$ columnar crystals, make the distribution of Nd-rich phase uniform and refine the grains of magnets, which results in the improvement of magnetic properties [14–18]. In order to further study the effect of heavy rare earth metals on the microstructure, phase formation, phase transition and magnetic properties of Nd-Fe-B-based permanent magnets, phase equilibria and thermodynamic information of the RE-Fe-B alloys are essential, which is useful to design novel multi-component RE-Fe-B magnetic alloys.

In order to establish the thermodynamic database of multi-component RE-Fe-B alloy systems, the Fe-RE and RE-B binary systems as the key sub-binary systems are fundamental. The Fe-RE binary systems such as Fe-La, Fe-Ce, Fe-Pr, Fe-Nd, Fe-Sm, Fe-Gd, Fe-Dy, Fe-Tb, Fe-Ho and Fe-Tm have been assessed in the literature [19–21] and in our previous work [22–25]. On the other hand, the RE-B binary systems such as La-B, Ce-B, Pr-B, Nd-B, Er-B, Dy-B and Lu-B were optimized in our previous work [26–28] and in the literature [29–34]. The Er-B binary system was optimized by Wang et al. [33] and Wang et al. [34]. The calculated results [33,34] are in agreement with the experimental data [35] and the reported results [36–38]. However, the calculated phase boundary of liquid phase in the composition ranges of 10-20 and 80-95 at.% B [33] is obvious deviation with the experimental data [35]. The calculated enthalpy of formation of intermetallic compound ErB_2 [34] is much more negative than the experimental data [39] and the calculated values [40–42]. In addition, the Ho-B and Tm-B binary systems were not assessed in the literature.

Therefore, thermodynamic assessment of the RE-B (Ho-B, Er-B, Tm-B) binary systems was performed in this work through the CALPHAD method on the basis of the critical assessment of phase diagram data and thermodynamic information in the published literature.

2. Literature information

2.1. The Ho-B binary system

Phase equilibria of the Ho-B binary system was reported by Spear

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[37,38]. There are five intermetallic compounds (HoB_2, HoB_4, HoB_6, HoB_{12} and HoB_{66}) and three solution phases (liquid, hcp(α-Ho), rhombohedral(β-B)). The crystal structures of five intermetallic compounds were determined by Roger [43] using X-ray diffraction, scanning electron microscopy and electron probe microanalysis. Table 1 shows the crystallographic information of intermetallic compounds in the Ho-B binary system. According to the reported results [37,38], four intermetallic compounds, HoB_2, HoB_6, HoB_{12} and HoB_{66}, were formed by four peritectic reactions (L + HoB_4 ↔ HoB_2 at 2473 K, L + HoB_4 ↔ HoB_6 at 2453 K, L + HoB_6 ↔ HoB_{12} at 2373 K and L + HoB_{12} ↔ HoB_{66} at 2298 K). The intermetallic compound HoB_4 was formed by the congruent reaction (L ↔ HoB_4 at 2773 K). Two eutectic reactions (L ↔ α-Ho + HoB_2 at 1723 K and L ↔ HoB_{66} + β-B at 2273 K) occur. The solubility of B in α-Ho or Ho in β-B was not reported in the literature.

The enthalpy of formation of HoB_2 was measured by Meschel and Kleppa [44] to be -27.9 ± 1.5 kJ/mole-atom by the means of direct synthesis calorimetry, and was calculated by Colinet and Tedenac [40] to be -57.4 kJ/mole-atom using the first principle calculation. Moreover, the enthalpies of formation of HoB_2, HoB_4, HoB_6 and HoB_{12} were calculated by the Open Quantum Materials Database (OQMD) [41] to be -57.02 kJ/mole-atom, -57.11 kJ/mole-atom, -36.92 kJ/mole-atom and -24.23 kJ/mole-atom, respectively, while were also calculated to be -57.31 kJ/mole-atom, -57.31 kJ/mole-atom, -36.92 kJ/mole-atom and -24.13 kJ/mole-atom by the Materials Project [42]. The enthalpies of formation of intermetallic compounds in the Ho-B binary system mentioned above were shown in Table 2.

2.2. The Er-B binary system

The phase diagram of the Er-B binary system was determined experimentally by Impearo [35] by means of differential thermal analysis, X-ray powder diffraction and electron microscope and then was reviewed by Liao and Spear [36] and Spear [37,38]. Four intermetallic compounds, ErB_2, ErB_4, ErB_{12} and ErB_{66}, are stable at room temperature. The crystal structures of ErB_2, ErB_4, ErB_{12} and ErB_{66} was studied by Castellano [45], Fisk et al. [46], LaPlaca et al. [47] and Spear and Solovyev [48] using X-ray diffraction analysis, respectively. Table 1 shows the crystallographic information of the stable phases in the Er-B binary system. On the basis of the experimental results [35] and the reported results [36–38], the intermetallic compounds ErB_2 and ErB_{12} were formed by two peritectic reactions (L + ErB_4 ↔ ErB_2 at 2458 K and L + ErB_4 ↔ ErB_{12} at 2356 K), while two intermetallic compounds ErB_4 and ErB_{66} were formed by two congruent reactions (L ↔ ErB_4 at 2633 K

Table 1
Crystallographic information of stable phases in the RE-B (RE=Ho, Er, Tm) binary system.

Phase	Prototype	Pearson symbol	Space group	Reference
α-Ho	Mg	hP2	$P6_3mmc$	[43]
HoB_2	AlB_2	hP3	P6/mmm	[43]
HoB_4	ThB_4	tP20	P4/mbm	[43]
HoB_6	CaB_6	cP7	$Pm\bar{3}m$	[43]
HoB_{12}	UB_{12}	cF52	$Fm\bar{3}m$	[43]
HoB_{66}	ThB_{66}	cF1880	$Fm\bar{3}c$	[43]
α-Er	Mg	hP2	$P6_3mmc$	[45]
ErB_2	AlB_2	hP3	P6/mmm	[45]
ErB_4	ThB_4	tP20	P4/mbm	[46]
ErB_{12}	UB_{12}	cF52	$Fm\bar{3}m$	[47]
ErB_{66}	ThB_{66}	cF1880	$Fm\bar{3}c$	[48]
α-Tm	Mg	hP2	$P6_3mmc$	[45]
TmB_2	AlB_2	hP3	P6/mmm	[45]
TmB_4	ThB_4	tP20	P4/mbm	[46]
TmB_{12}	UB_{12}	cF52	$Fm\bar{3}m$	[48]
TmB_{66}	ThB_{66}	cF1880	$Fm\bar{3}c$	[39]
β-B	β-B	hR105	$R\bar{3}m$	[51]

Table 2
The experimental and calculated enthalpies of formation of intermetallic compounds in the RE-B (RE=Ho, Er, Tm) binary systems.

Compounds	Enthalpies of formation (kJ/mol-atom)	Method	Reference
HoB_2	-27.9 ± 1.5	direct synthesis calorimetry	[44]
	-57.40	first-principle calculation	[40]
	-57.02	first-principle calculation	[41]
	-57.31	first-principle calculation	[42]
	-58.07	Calphad calculation	This work
HoB_4	-57.11	first-principle calculation	[41]
	-57.31	first-principle calculation	[42]
	-57.00	Calphad calculation	This work
HoB_6	-36.92	first-principle calculation	[41]
	-36.92	first-principle calculation	[42]
	-55.00	Calphad calculation	This work
HoB_{12}	-24.23	first-principle calculation	[41]
	-24.13	first-principle calculation	[42]
	-44.00	Calphad calculation	This work
HoB_{66}	-25.00	Calphad calculation	This work
ErB_2	-27.70	Calphad calculation	[33]
	-74.10	Calphad calculation	[34]
	-60.26	first-principle calculation	[40]
	-59.81	first-principle calculation	[41]
	-60.00	first-principle calculation	[42]
	-27.3 ± 2.2	direct synthesis calorimetry	[39]
	-59.30	Calphad calculation	This work
ErB_4	-17.10	Calphad calculation	[33]
	-54.02	Calphad calculation	[34]
	-56.42	first-principle calculation	[40]
	-56.54	first-principle calculation	[41]
	-56.63	first-principle calculation	[42]
	-57.73	Calphad calculation	This work
ErB_{12}	-6.60	Calphad calculation	[33]
	-32.13	Calphad calculation	[34]
	-24.12	first-principle calculation	[40]
	-24.23	first-principle calculation	[41]
	-24.23	first-principle calculation	[42]
	-29.78	Calphad calculation	This work
ErB_{66}	-1.30	Calphad calculation	[33]
	-12.42	Calphad calculation	[34]
	-7.01	Calphad calculation	This work
TmB_2	-63.06	first-principle calculation	[40]
	-62.60	first-principle calculation	[41]
	-62.79	first-principle calculation	[42]
	-30.6 ± 3	direct synthesis calorimetry	[49]
	-62.01	Calphad calculation	This work
TmB_4	-55.77		[41]

(continued on next page)

Table 2 (*continued*)

Compounds	Enthalpies of formation (kJ/mol-atom)	Method	Reference
		first-principle calculation	
	-55.19	first-principle calculation	[42]
	-55.00	Calphad calculation	This work
TmB_{12}	-24.23	first-principle calculation	[41]
	-24.13	first-principle calculation	[42]
	-26.93	Calphad calculation	This work
TmB_{66}	-9.97	Calphad calculation	This work

and L↔ ErB_{66} at 2343 K). There are three eutectic reactions (L ↔ α-Er + ErB_2 at 1773 K, L ↔ ErB_{12} + ErB_{66} at 2288 K, L ↔ ErB_{66} + β-B at 2293 K) in the Er-B binary system. The solubility of B in α-Er or Er in β-B was not published in the literature.

Using direct synthesis calorimetry, the enthalpy of formation of ErB_2 was measured by Meschel and Kleppa [48] to be -27.3 ± 2.2 kJ/mole-atom, while the enthalpies of formation of ErB_2, ErB_4 and ErB_{12} were calculated by Colinet and Tedenac [40] to be -60.26 kJ/mole-atom, -56.42 kJ/mole-atom and -24.12 kJ/mole-atom, respectively. Furthermore, the enthalpies of formation of ErB_2, ErB_4 and ErB_{12} were calculated by the Open Quantum Materials Database (OQMD) [41] to -59.81 kJ/mole-atom, -56.54 kJ/mole-atom and -24.23 kJ/mole-atom, respectively, while were also calculated by the Materials Project [42] to be -60.00 kJ/mole-atom, -56.63 kJ/mole-atom and -24.23 kJ/mole-atom. Table 2 gives the enthalpies of formation of intermetallic compounds in the Er-B binary system in the reported literature.

2.3. The Tm-B binary system

The Tm-B binary system was only reported by Spear [37,38]. The Tm-B binary system has four intermetallic compounds (TmB_2, TmB_4, TmB_{12} and TmB_{66}) and three solution phases (liquid, hcp(α-Tm) and rhombohedral(β-B)). The crystal structure of TmB_2 was studied by Castellano [45] through X-ray diffraction analysis, while those of TmB_4, TmB_{12} and TmB_{66} were determined by Fisk et al. [46], LaPlaca et al. [47] and Spear and Solovyev [48], respectively. Table 1 shows the crystallographic information of stable phases in the Tm-B binary system. According to the reported results [37,38], the intermetallic compounds TmB_2, TmB_{12} and TmB_{66} were formed by three peritectic reactions (L + TmB_4 ↔ TmB_2 at 2523 K, L + TmB_4 ↔ TmB_{12} at 2453 K, L + TmB_{12} ↔ TmB_{66} at 2373 K), while the intermetallic compound TmB_4 melts congruently at 2823 K. The temperature of the eutectic reaction (L ↔ α-Tm + TmB_2) was reported to be 1773 K, while the temperature of the eutectic reaction (L ↔ TmB_{66} + β-B) was not given. The solubility of B in α-Tm or Tm in β-B was not studied in the literature.

The enthalpy of formation of TmB_2 was determined by Meschel et al. [49] to be -30.6 ± 3 kJ/mole-atom by direct synthesis calorimetry. Using the first principle calculation, the enthalpy of formation of TmB_2 calculated by Colinet et al. [40] is -63.06 kJ/mole-atom. The enthalpies of formation of TmB_2, TmB_4 and TmB_{12} were calculated by the Open Quantum Materials Database (OQMD) [41] to -62.60 kJ/mole-atom, -55.77 kJ/mole-atom and -24.23 kJ/mole-atom, respectively, while were also calculated by the Materials Project [42] to be -62.79 kJ/mole-atom, -55.19 kJ/mole-atom and -24.13 kJ/mole-atom. Table 2 shows the enthalpies of formation of intermetallic compounds in the Tm-B binary system in the reported literature.

3. Thermodynamic modeling

3.1. Solution phases

In the RE-B (RE=Ho, Er, Tm) binary system, the molar Gibbs energy of solution phase φ (e.g. liquid, hcp(α-Ho, α-Er, α-Tm) and rhombohedral(β-B)) is expressed by the substitutional solution model as follows:

$$G_m^{\varphi} = x_{RE}{}^0G_{RE}^{\varphi} + x_B{}^0G_B^{\varphi} + RT(x_{RE}\ln x_{RE} + x_B\ln x_B) + {}^{ex}G_m^{\varphi} \quad (1)$$

$${}^0G_i^{\varphi} = G_i^{\varphi}(T) - H_i^{SER}(298.15) = a + bT + cT\ln T + dT^2 + eT^{-1} + fT^3 + gT^7 + hT^{-9} \quad (2)$$

$${}^{ex}G_m^{\varphi} = x_{RE}x_B\sum_{n=0}{}^nL_{RE,B}^{\varphi}(x_B - x_{RE})^n \quad (3)$$

$${}^nL_{RE,B}^{\varphi} = A_n + B_nT \quad (4)$$

where ${}^0G_B^{\varphi}$ and ${}^0G_{RE}^{\varphi}$ are the molar Gibbs energies of pure elements B and RE (RE=Ho, Er, Tm) with the structure φ in the reference state (298.15 K, 1 bar) taken from the SGTE (Scientific Group Thermodata Europe) Compiled by Dinsdale [50]. x_B xRE and x_{RE} xBare mole fractions of the pure element B and RE (RE=Ho, Er, Tm), respectively. R is the gas constant and T is the absolute temperature in Kelvin. ${}^{ex}G_m^{\varphi}$ is the excess Gibbs energy of the phase φ described by the Redlich-Kister polynomial. ${}^nL_{RE,B}^{\varphi}$ is the interaction parameter dependent on the temperature as expressed by two constants A_n and B_n to be optimized in this work.

It is noticed that considering no experimental solubility of B in hcp (α-Ho, α-Er, α-Tm) phases and RE in rhombohedral(β-B) phases, these solid phases are regarded as the pure component phases in this work.

3.2. Intermetallic compounds

In the RE-B (RE=Ho, Er, Tm) binary system, all intermetallic compounds (HoB_2, HoB_4, HoB_6, HoB_{12}, HoB_{66}, ErB_2, ErB_4, ErB_{12}, ErB_{66}, TmB_2, TmB_4, TmB_{12} and TmB_{66}) are treated as stoichiometric compound because of no available experimental information for their composition ranges in the literature. Due to lack of the experimental heat capacity, the molar Gibbs energies of intermetallic compounds RE_pB_q are expressed using the Neumann-Kopp rule as:

$$G_m^{RE_pB_q} = \frac{p}{p+q}{}^0G_{RE}^{hcp} + \frac{q}{p+q}{}^0G_B^{rho} + J + KT \quad (5)$$

where p and q are the values of stoichiometric numbers as well as J and K are the parameters to be optimized in this work.

4. Calculated results

Using the PARROT module of the Thermo-Calc® software [52], thermodynamic parameters of the phases in the RE-B (RE=Ho, Er, Tm) binary systems were strictly optimized on the basis of the experimental information and the reported results in the literature. The parameters of three Ho-B, Er-B and Tm-B binary systems obtained finally in this work were given in Table 3, Table 4 and Table 5, respectively. The calculated results were shown in the following section.

4.1. The Ho-B binary system

Fig. 1 shows the calculated Ho-B binary phase diagram with the reported data [37,38]. As can be seen, the calculated Ho-B phase diagram in this work are well consistent with the reported data by Spear [37,38]. Table 6 shows the calculated invariant reactions in the Ho-B binary system with the reported values [37,38]. In this work, the calculated temperatures of three peritectic reactions (L + HoB_4 ↔ HoB_2, L + HoB_4

Table 3
Thermodynamic parameters of the Ho-B binary system.

Phase	Thermodynamic Parameters	Reference
liquid	${}^{0}L^{liq}_{B,Ho} = -84000 + 17T$ ${}^{1}L^{liq}_{B,Ho} = -50000 + 6T$ ${}^{2}L^{liq}_{B,Ho} = -19994.674 + 2T$ ${}^{3}L^{liq}_{B,Ho} = -17500$	This work
hcp (α-Ho)	${}^{0}G^{hcp}_{Ho}$ cited from SGTE database	[50]
rhombohedral (β-B)	${}^{0}G^{rho}_{B}$ cited from SGTE database	[50]
HoB_2	$G^{HoB_2}_{m} = -58067.212 + 10.78T + 0.3333\,{}^{0}G^{hcp}_{Ho} + 0.6667\,{}^{0}G^{rho}_{B}$	This work
HoB_4	$G^{HoB_4}_{m} = -57000 + 9.48T + 0.2\,{}^{0}G^{hcp}_{Ho} + 0.8\,{}^{0}G^{rho}_{B}$	This work
HoB_6	$G^{HoB_6}_{m} = -55000 + 12.168T + 0.1429\,{}^{0}G^{hcp}_{Ho} + 0.8571\,{}^{0}G^{rho}_{B}$	This work
HoB_{12}	$G^{HoB_{12}}_{m} = -44000 + 12.39T + 0.0769\,{}^{0}G^{hcp}_{Ho} + 0.9231\,{}^{0}G^{rho}_{B}$	This work
HoB_{66}	$G^{HoB_{66}}_{m} = -25000 + 9.4T + 0.0149\,{}^{0}G^{hcp}_{Ho} + 0.9851\,{}^{0}G^{rho}_{B}$	This work

Table 4
Thermodynamic parameters of the Er-B binary system.

Phase	Thermodynamic Parameters	Reference
liquid	${}^{0}L^{liq}_{B,Er} = -87017.4 + 10.8615T$ ${}^{1}L^{liq}_{B,Er} = -64312.861 + 4T$ ${}^{2}L^{liq}_{B,Er} = -20002 + 2.0002T$ ${}^{3}L^{liq}_{B,Er} = -15503.1$	This work
hcp (α-Er)	${}^{0}G^{hcp}_{Er}$ cited from SGTE database	[50]
rhombohedral (β-B)	${}^{0}G^{rho}_{B}$ cited from SGTE database	[50]
ErB_2	$G^{ErB_2}_{m} = -59305.915 + 9.7459T + 0.3333\,{}^{0}G^{hcp}_{Er} + 0.6667\,{}^{0}G^{rho}_{B}$	This work
ErB_4	$G^{ErB_4}_{m} = -57726.603 + 9.0205T + 0.2\,{}^{0}G^{hcp}_{Er} + 0.8\,{}^{0}G^{rho}_{B}$	This work
ErB_{12}	$G^{ErB_{12}}_{m} = -29782.817 + 5.9904T + 0.0769\,{}^{0}G^{hcp}_{Er} + 0.9231\,{}^{0}G^{rho}_{B}$	This work
ErB_{66}	$G^{ErB_{66}}_{m} = -7012.902 + 1.3954T + 0.0149\,{}^{0}G^{hcp}_{Er} + 0.9851\,{}^{0}G^{rho}_{B}$	This work

↔ HoB_6 and L + HoB_6 ↔ HoB_{12}) and one congruent reaction (L ↔ HoB_4) are in good agreement with the reported values [37,38]. On the other hand, the calculated temperature of the eutectic reaction (L ↔ α-Ho + HoB_2) is lower (17 K) than the reported value [37,38]. The temperatures of two invariant reactions (L + HoB_{12} ↔ HoB_{66} and L ↔ HoB_{66} + β-B) are higher (41 K and 65 K) than the reported values [37,38], respectively. The calculated liquid compositions of three peritectic reactions (L + HoB_4 ↔ HoB_2, L +HoB_4 ↔ HoB_6 and L + HoB_6 ↔ HoB_{12}) are obvious deviation with the reported values by Spear [37,38]. The reasons for the deviation with the reported results [37,38] could be that the liquid boundary of rich-B part is very steep and HoB_6, HoB_{12} and HoB_{66} are formed by three peritectic reactions in the narrow composition range

Table 5
Thermodynamic parameters of the Tm-B binary system.

Phase	Thermodynamic Parameters	Reference
liquid	${}^{0}L^{liq}_{B,Tm} = -93000 + 18.1147T$ ${}^{1}L^{liq}_{B,Tm} = -53599.437 + 4.2952T$ ${}^{2}L^{liq}_{B,Tm} = -20171.147 - 1.9219T$ ${}^{3}L^{liq}_{B,Tm} = -17276.952$	This work
hcp (α-Tm)	${}^{0}G^{hcp}_{Tm}$ cited from SGTE database	[50]
rhombohedral (β-B)	${}^{0}G^{rho}_{B}$ cited from SGTE database	[50]
TmB_2	$G^{TmB_2}_{m} = -62010.006 + 10.9125T + 0.3333\,{}^{0}G^{hcp}_{Tm} + 0.6667\,{}^{0}G^{rho}_{B}$	This work
TmB_4	$G^{TmB_4}_{m} = -55000 + 7.23T + 0.2\,{}^{0}G^{hcp}_{Tm} + 0.8\,{}^{0}G^{rho}_{B}$	This work
TmB_{12}	$G^{TmB_{12}}_{m} = -26930.98 + 4.167T + 0.0769\,{}^{0}G^{hcp}_{Tm} + 0.9231\,{}^{0}G^{rho}_{B}$	This work
TmB_{66}	$G^{TmB_{66}}_{m} = -9968.801 + 2.495T + 0.0149\,{}^{0}G^{hcp}_{Tm} + 0.9851\,{}^{0}G^{rho}_{B}$	This work

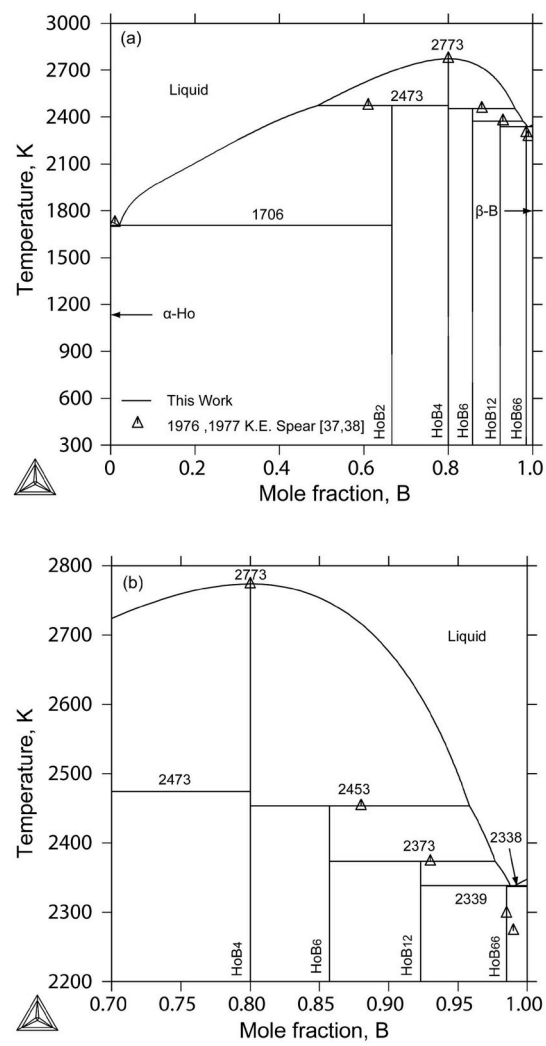

Fig. 1. The calculated Ho-B phase diagram in this work with the reported data [37,38].

Table 6
Invariant reactions in the Ho-B binary system.

Reactions	Reaction type	Composition (x_B^{liq})	Temperature (K)	Reference
L ↔ α-Ho + HoB_2	Eutectic	0.010 0.021	1723 1706	[37,38] This work
L + HoB_4 ↔ HoB_2	Peritectic	0.610 0.492	2473 2473	[37,38] This work
L ↔ HoB_4	Congruent	0.800 0.800	2773 2773	[37,38] This work
L + HoB_4 ↔ HoB_6	Peritectic	0.880 0.959	2453 2453	[37,38] This work
L + HoB_6 ↔ HoB_{12}	Peritectic	0.930 0.977	2373 2373	[37,38] This work
L + HoB_{12} ↔ HoB_{66}	Peritectic	0.985 0.987	2298 2339	[37,38] This work
L ↔ HoB_{66} + β-B	Eutectic	0.990 0.991	2273 2338	[37,38] This work

(85-98 at.% B). The compositions of liquid phase for three peritectic reactions and one eutectic reaction (L ↔ HoB_{66} + β-B) are very close to the compositions of HoB_6, HoB_{12} and HoB_{66} compounds. Moreover, the liquid boundary of HoB_4 formed by the congruent reaction is not symmetrical. Thus, further experiments should be performed to determine accurately phase equilibira of the Ho-B binary system.

Fig. 2 is the calculated enthalpies of formation of intermetallic compounds in the Ho-B binary system at 298 K. The calculated enthalpies of formation of intermetallic compounds in this work agree well with the calculated results [40–42]. However, the calculated enthalpy of formation of HoB_2 in this work and in Refs. [40–42] is much more negative than the experimental data measured by Meschel and Kleppa [44].

4.2. The Er-B binary system

Fig. 3 compares the calculated Er-B binary phase diagram in this work with the calculated results [33,34]. Fig. 4 is the calculated results in this work with the experimental data [35]. The calculated phase boundary of liquid phase in this work agrees well with the experimental data [35]. In particular, the calculated liquidus in the Er-rich part is in much better agreement with the experimental results than the

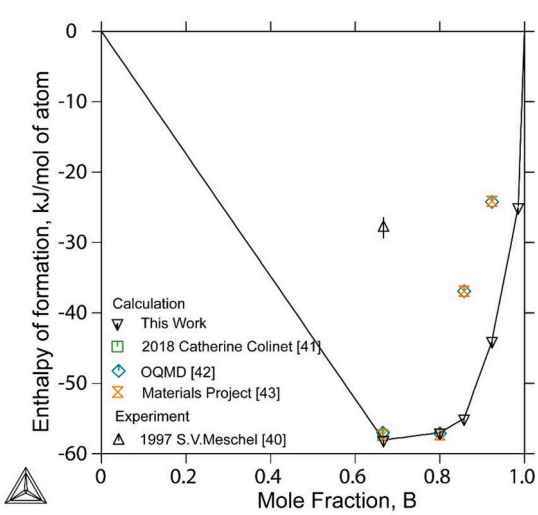

Fig. 2. The calculated enthalpies of formation of Ho-B intermetallic compounds at 298 K with the experimental data [44] and the calculated results [40–42]. (Reference states: α-Ho and β-B).

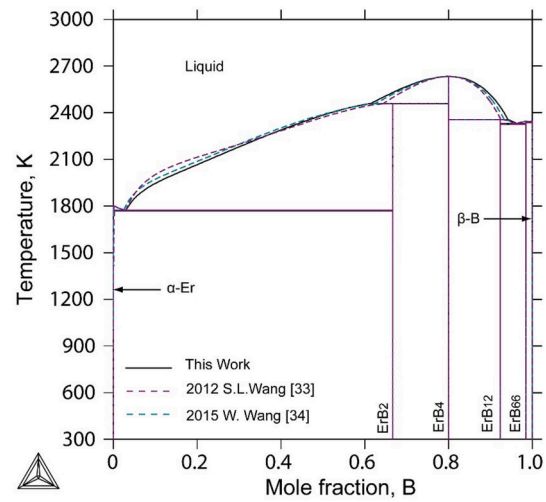

Fig. 3. The calculated Er-B phase diagram in this work with the calculated results [33,34].

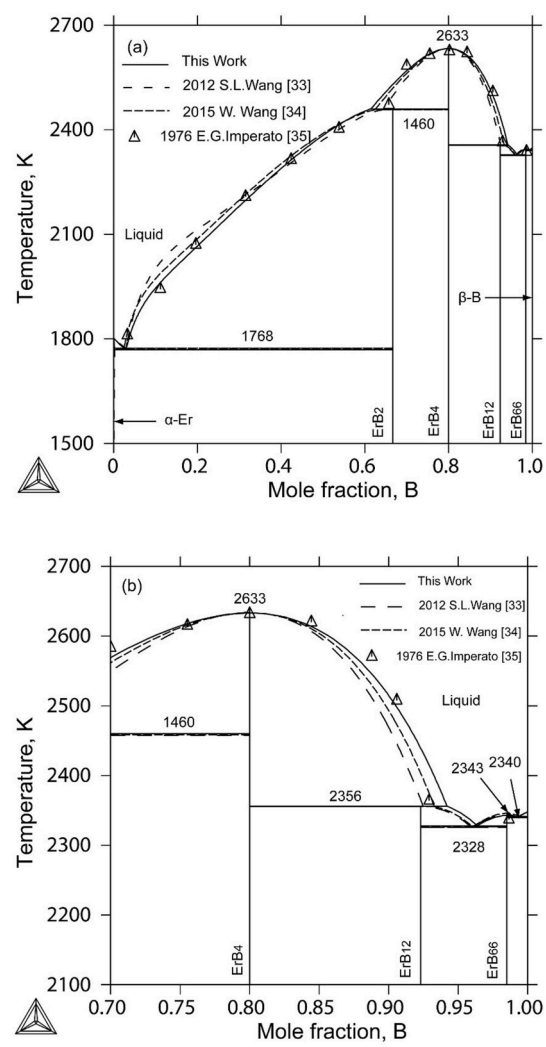

Fig. 4. The calculated Er-B phase diagram in this work with the experimental data [35] and the calculated results [33,34].

calculated results [33,34]. Table 7 shows the calculated invariant reactions in this work with the previously calculated results [33,34] and the reported values [36–38]. In this work, the calculated temperatures of two peritectic reactions (($L + ErB_4 \leftrightarrow ErB_2$ and $L + ErB_4 \leftrightarrow ErB_{12}$), two congruent reactions ($L \leftrightarrow ErB_4$ and $L \leftrightarrow ErB_{66}$) and one eutectic reaction (($L \leftrightarrow \alpha\text{-Er} + ErB_2$) agree well with the experimental data [35] and the previously calculated results [33,34]. The calculated temperatures of two invariant reactions ($L \leftrightarrow ErB_{12} + ErB_{66}$ and $L \leftrightarrow ErB_{66} + \beta\text{-B}$) are consistent with the previously calculated results [33,34] and are higher (40 K and 47 K) than the experimental data [35]. In addition, the liquid compositions of two peritectic reactions ($L + ErB_4 \leftrightarrow ErB_2$ and $L + ErB_4 \leftrightarrow ErB_{12}$) are not well consistent with the experimental data [35]. The reasons for the deviation are similar with the results of the Ho-B binary system mentioned above.

Fig. 5 displays the calculated enthalpies of formation of intermetallic compounds ErB_2, ErB_4, ErB_{12} and ErB_{66} at 298 K with the experimental data [39] and the calculated results [33,34,40–42]. The calculated results in this work are consistent with the calculated results [40–42], and are obvious deviation with the previously calculated results [33,34]. Especially, the calculated enthalpy of formation of ErB_2 in this work is much more negative than the experimental data measured by Meschel et al. [39] and the calculated results [33].

4.3. The Tm-B binary system

Fig. 6 is the calculated Tm-B binary phase diagram with the reported results [37,38]. Table 8 gives the calculated temperatures of invariant reactions with the reported results [37,38]. The calculated temperatures of three peritectic reactions ($L + TmB_4 \leftrightarrow TmB_2$, $L + TmB_4 \leftrightarrow TmB_{12}$ and $L + TmB_{12} \leftrightarrow TmB_{66}$), one eutectic reaction ($L \leftrightarrow \alpha\text{-Tm} + TmB_2$) and one congruent reaction ($L \leftrightarrow TmB_4$) are in good agreement with the reported values [37,38]. It was noticed that the temperature of the eutectic reaction ($L \leftrightarrow TmB_{66} + \beta\text{-B}$) was not reported in the literature.

Fig. 7 is the calculated enthalpies of formation of intermetallic compounds in the Tm-B binary system at 298 K with the experimental data [49] and the calculated results [40–42]. In this work, the calculated enthalpies of formation of intermetallic compounds are in good

Table 7
Invariant reactions in the Er-B binary system.

Reactions	Reaction type	Composition (x_B^{liq})	Temperature (K)	Reference
$L \leftrightarrow \alpha\text{-Er} + ErB_2$	Eutectic	~0.010	1773	[35]
		0.029	1770	[33]
		0.025	1773	[34]
		0.030	1768	This work
$L + ErB_4 \leftrightarrow ErB_2$	Peritectic	0.650	2458	[35]
		0.643	2457	[33]
		0.644	2458	[34]
		0.615	2460	This work
$L \leftrightarrow ErB_4$	Congruent	0.800	2633	[35]
		0.800	2633	[33]
		0.800	2633	[34]
		0.800	2633	This work
$L + ErB_4 \leftrightarrow ErB_{12}$	Peritectic	0.930	2356	[35]
		0.925	2356	[33]
		0.926	2356	[34]
		0.942	2356	This work
$L \leftrightarrow ErB_{12} + ErB_{66}$	Eutectic	0.970	2288	[35]
		0.960	2325	[33]
		0.960	2325	[34]
		0.963	2328	This work
$L \leftrightarrow ErB_{66}$	Congruent	0.985	2343	[35]
		0.985	2343	[33]
		0.985	2343	[34]
		0.985	2343	This work
$L \leftrightarrow ErB_{66} + \beta\text{-B}$	Eutectic	~0.990	2293	[35]
		0.992	2340	[33]
		0.992	2340	[34]
		0.992	2340	This work

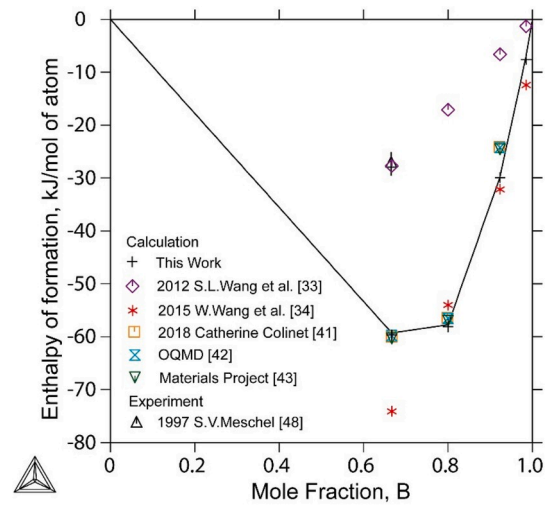

Fig. 5. The calculated enthalpies of formation of Er-B intermetallic compounds at 298 K with the experimental data [39] and the calculated results [33,34, 40–42]. (Reference states: α-Er and β-B).

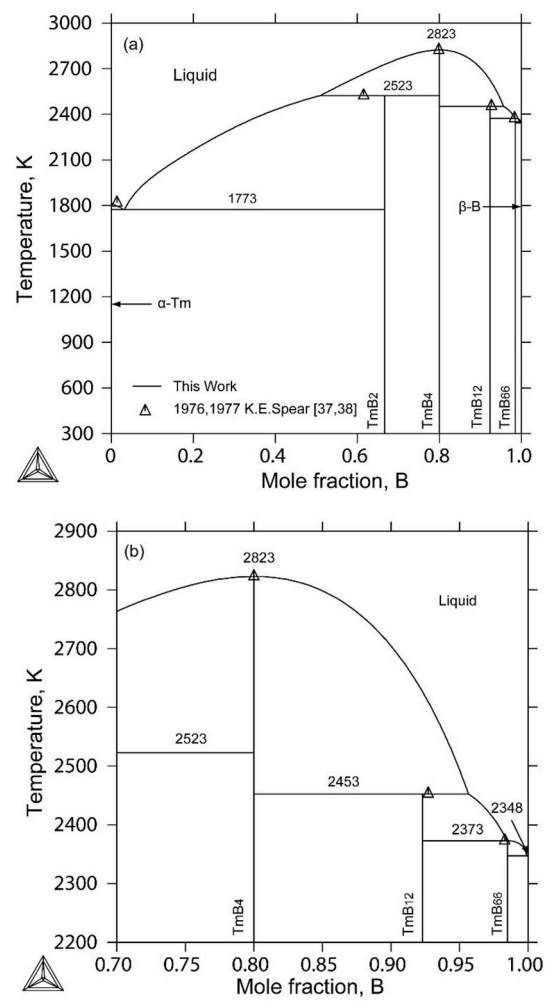

Fig. 6. The calculated Tm-B phase diagram in this work with the reported data [37,38].

Table 8
Invariant reactions in the Tm-B binary system.

Reactions	Reaction type	Composition (x_B^{liq})	Temperature (K)	Reference
L ↔ α-Tm + TmB_2	Eutectic	–	1773	[37,38]
		0.033	1773	This work
L ↔ TmB_4	Congruent	–	2823	[37,38]
		0.800	2823	This work
L + TmB_4 ↔ TmB_2	Peritectic	–	2523	[37,38]
		0.510	2523	This work
L + TmB_4 ↔ TmB_{12}	Peritectic	–	2453	[37,38]
		0.957	2453	This work
L + TmB_{12} ↔ Tm_{66}	Peritectic	–	2373	[37,38]
		0.985	2373	This work
L ↔ TmB_{66} + β-B	Eutectic	0.999	2348	This work

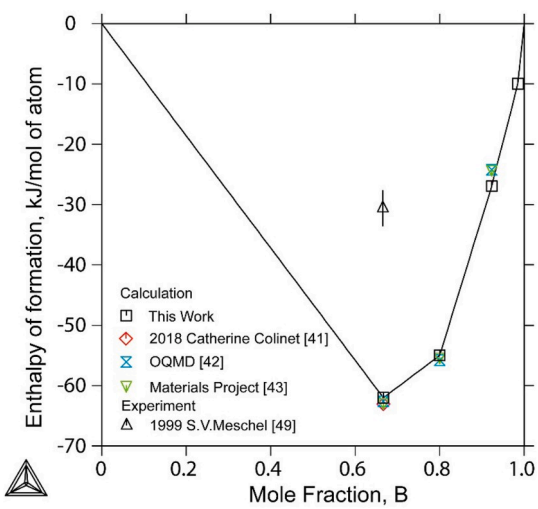

Fig. 7. The calculated enthalpies of formation of Tm-B intermetallic compounds at 298 K with the experimental data [49] and the calculated results [40–42]. (Reference states: α-Tm and β-B).

agreement with the calculated results [40–42]. Meanwhile, the calculated enthalpy of formation of TmB_2 is much more negative than the experimental data determined by Meschel and Kleppa [49].

5. Discussion

As shown in the section of literature information, it was found that the experimental results including phase diagram data and thermodynamic properties in the RE-B (RE=Ho, Er, Tm) binary systems are limited due to the experimental difficulties (e.g. high temperature and oxidization). In this work, the calculated phase diagrams of the RE-B (RE=Ho, Er, Tm) binary systems agree well, although the calculated temperatures and compositions of some invariant reactions shows a little deviation with the experimental data [35] and the reported results [37,38]. The calculated enthalpies of formation of intermetallic compounds in the RE-B (RE=Ho, Er, Tm) binary systems are consistent with the first principle calculation results [40–42]. In order to clarify the validity of the present calculations, phase equilibria and thermodynamic information of the RE-B binary systems can be analyzed relatively considering similar physical and chemical properties of rare earth metals.

According to the reported information, in general, the RE-B binary system has five stable intermetallic compounds, REB_2, REB_4, REB_6, REB_{12} and REB_{66}. The Ho-B binary system has five intermetallic compounds, while the Er-B and Tm-B binary systems have only four intermetallic compounds (REB_2, REB_4, REB_{12} and REB_{66}). The intermetallic compounds REB_2 and REB_4 are formed by the eutectic reaction and the congruent reaction, respectively, while the intermetallic compounds REB_6 and REB_{12} are formed by peritectic reactions. The intermetallic compound ErB_{66} is formed by the congruent reaction, while the intermetallic compounds HoB_{66} and TmB_{66} are formed by the peritectic reaction. Furthermore, the temperatures of formation of HoB_2, HoB_4 and HoB_{12} are lower, while those of formation of TmB_2, TmB_4 and TmB_{12} are higher. The temperatures of formation of REB_{66} become higher with increasing of RE atomic numbers.

On the other hand, the experimental and calculated enthalpies of formation of intermetallic compounds in the RE-B (RE=Ho, Er, Tm) binary system were summarized in Table 2. The calculated enthalpies of formation of intermetallic compounds REB_2 show the deviation with the experimental data [39,44,49], although the calculated enthalpies of formation of all intermetallic compounds are consistent with the first principle calculation results [40–42]. It is very evident that the enthalpies of formation of intermetallic compounds in the RE-B binary systems become much more negative with the increase of RE atomic numbers, while those of the intermetallic compounds REB_2 is most negative. The similar results were given by Li et al. [32] in the calculation of the enthalpy of formation of intermetallic compounds REB_2 in the Ce-B, Dy-B and Lu-B binary systems. Fig. 8 shows the calculated enthalpies of formation of RE-B (RE=Dy, Ho, Er, Tm, Lu) intermetallic compounds at 298 K with the experimental data [39,44,49] and the calculated results [32]. According to the calculated results [32] and in the present work, it is obvious that the calculated enthalpies of formation of RE-B (RE=Dy, Ho, Er, Tm, Lu) intermetallic compounds by the first principle calculation match well with each other, and show obvious deviation with the experimental data [39,44,49].

Fig. 9 shows the calculated enthalpies of mixing of liquid RE-B (RE=Ho, Er, Tm) alloys at 3000 K. In order to analyze systematically the mixing enthalpy of liquid RE-B alloys, the mixing enthalpies of liquid Dy-B and Lu-B alloys calculated by Li et al. [32] were also given in Fig. 9. It can be seen that the calculated enthalpies of mixing of liquid RE-B (RE=Dy, Ho, Er, Tm, Lu) alloys are negative deviation, and show most minimum values in the composition range of 60-70% at.% B. With the increase of RE atomic numbers, the calculated enthalpies of mixing of liquid RE-B (RE=Dy, Ho, Er, Tm, Lu) alloys become much more negative. This trend is consistent with the change of the formation enthalpy of intermetallic compounds in the RE-B (RE=Dy, Ho, Er, Tm, Lu) binary

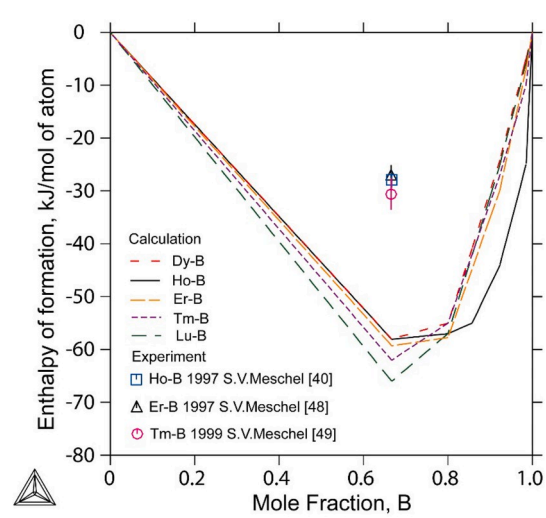

Fig. 8. The calculated enthalpies of formation of RE-B (RE=Dy, Ho, Er, Tm, Lu) intermetallic compounds at 298 K with the experimental data [38,44,49] and the calculated results [32]. (Reference states: α-Dy, α-Ho, α-Er, α-Tm, α-Lu and β-B).

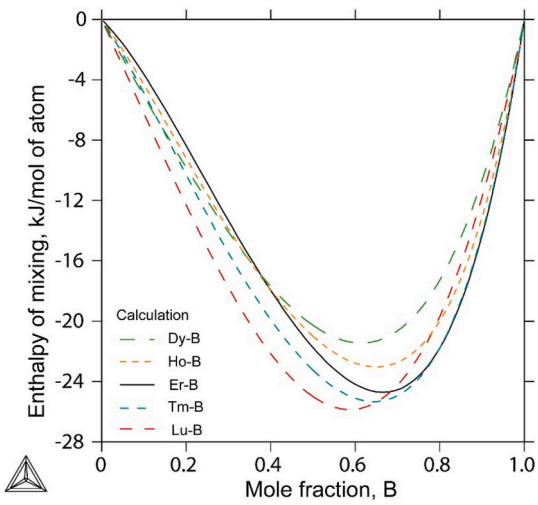

Fig. 9. The calculated enthalpies of mixing of liquid RE-B (RE=Dy, Ho, Er, Tm, Lu) alloys at 3000 K in this work. (Reference states: liquid RE and liquid B).

systems. Therefore, experimental investigations are still necessary to be performed due to lack of experimental information including mixing enthalpy and formation enthalpy in the RE-B binary systems.

6. Conclusions

The RE-B (RE=Ho, Er, Tm) binary systems were assessed thermodynamically using the CALPHAD method on the basis of the experimental information reported in the literature. Self-consistent thermodynamic parameters of the RE-B (RE=Ho, Er, Tm) binary systems were achieved finally. The calculated results are in good agreement with the reported results. Phase diagrams and thermodynamic properties of the RE-B (RE=Ho, Er, Tm) binary systems was discussed considering the similarly physical and chemical properties of rare earth metals.

Declaration of competing interest

The authors of the manuscript 'Thermodynamic assessment of the RE-B (RE=Ho, Er, Tm) binary systems' declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2020.101796.

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